

Civic Loops: Regulating the Feedback Structure, Not the Knob

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August 5, 2026

Abstract

Two companions prove that satisfaction pressure selects harmful deference and satisfaction ascent drifts the knob into capture, where the harm is invisible to the optimized metric. This paper moves from the knob to the closed loop it lives inside: population beliefs as plant, a satisfaction-ascending operator as controller, a content channel feeding contested claims at gain g from standing disagreement, and a measurement channel from error to observable. Five results follow. An aggregation–conservation theorem locates the companions’ knife edge in a complete classification: stationary measured agreement is conserved on two exact loci, rewards capture below the grounding rate λ_0 , and admits an interior partial-capture optimum above it. A knob-free certificate, $v(\lambda_0 - g) \geq 2cI$, forces every sign-consistent adaptation rule at any rate to calibration; below the capture line the corner is stable, with stable partial-capture equilibria between. Gain ceilings are exact – captured stocks diverge precisely when gain exceeds grounding – and the certificate becomes componentwise on networks under $\rho(\bar{S} + G) < 1$. Laundering becomes exact nonlinear unobservability: at capture the satisfaction output depends on the healthy coordinate alone, and an ε -weighted audit sensor restores detectability with $\ln(1/\varepsilon)$ latency. Rescaled transit converges to an α -free jump map whose closed-form threshold scale governs the capture separatrix. Loop parameters are identifiable from contested-stream panels, the estimated certificate predicts fate out of sample, and a safe-side rule certifies no capturable loop in simulation. The regulatory translation is a menu of inequalities – gain caps, grounding floors, audit-sensor mandates, cadence caps – none of which mentions the knob. Every theorem is machine-checked in Lean 4, every claim battery-verified.

1 Introduction: the change of object

The first paper in this sequence ([Hofheinz, 2026b](#)) treats the deference parameter β as chosen once and proves that per-interaction satisfaction pressure selects $\beta^* > 0$ at a strict cost to collective epistemic capacity. The second ([Hofheinz, 2026c](#)) makes the pressure dynamical: an operator running gradient ascent on measured satisfaction, against a disagreement stock that deference itself replenishes, is a planar cooperative map that is bistable between a calibrated and a captured regime; crossing the basin boundary is an accumulative risk in the sense of [Kasirzadeh \(2025\)](#), and at full capture measured satisfaction becomes structurally blind to the harm. That last result has a corrosive implication for governance: any regulation phrased in terms of β , or of satisfaction telemetry, regulates a quantity that is endogenous to the loop being optimized – and the loop launders. The deference knob is the wrong object.

This paper develops the mathematics of the right object: the closed loop itself. The loop has four parts. The *plant* is the population’s portfolio of contested claims, carrying a disagreement stock

D. The *controller* is the operator, adapting β by some rule that ascends measured satisfaction. The *content channel* is new in this paper: standing disagreement shapes the arrival of fresh contested claims – polarized environments generate polarized questions – at a gain $g \geq 0$ per unit stock. The *measurement channel* maps population error into the operator’s observable, weighting each user’s error by that user’s residual independence; it is this weighting that launders. The empirical record motivating the controller and the channels is the same as the companions’: preference models measurably reward sycophantic agreement (Sharma et al., 2024), over-optimizing learned satisfaction proxies degrades true objectives (Gao et al., 2023), RLHF can teach models to mislead evaluators rather than inform them (Wen et al., 2025)¹, and optimizing for user feedback discovers targeted manipulation (Williams et al., 2025).

The geometric heart of the change of object is already visible in the companion’s phase portrait. Its two nullclines are the input–output characteristics of the two subsystems – the population’s stationary stock $D^*(\beta)$ and the operator’s best response $\hat{\beta}(D)$. The knob moves the *state along* the curves. Every regulatory instrument worth the name moves the *curves*. Safety is a property of the curve pair, and because the loop is monotone, that property is certifiable by the sign of a single scalar function, in the spirit of the small-gain theory of monotone input/output systems (Angeli and Sontag, 2003, 2004); the multistability of the companion is exactly the crossing-pattern multistability of that theory. The dynamical backbone is the convergence theory of planar cooperative maps (Smith, 1998; Hirsch and Smith, 2006); the observability language is classical (Sontag, 1998); the rate-separation analysis is the deterministic skeleton of two-timescale stochastic approximation (Borkar, 1997, 2008); and the network extension inherits the opinion-dynamics lineage of DeGroot (1974), Friedkin and Johnsen (1990), and Golub and Jackson (2010). The content channel has recommender-system antecedents – algorithmic confounding and degenerate feedback loops, the algorithm reshaping the data it trains on (Chaney et al., 2018; Jiang et al., 2019), with induced preference shift the operator-side hazard (Carroll et al., 2022) – and the interconnection is performative in the sense of Perdomo et al. (2020): the observable the controller ascends is measured against a plant its own policy moves. What the loop framing adds is what those lines do not carry: certificates, ceilings, and observability structure for the crossing pattern, rather than stability of a single operating point.

Section 2 defines the loop and proves the classification theorem: one step-size condition makes the loop a monotone map, a strictly stronger one makes every orbit converge to an equilibrium, and the equilibrium set with stabilities is read off the sign pattern of one scalar function G_g . Section 3 proves the aggregation–conservation theorem, a complete map of which (stock, loss, feedback) triples conserve stationary measured agreement, which reward capture, and which exhibit interior partial-capture optima. Section 4 proves the knob-free certificate and the zone structure, including the existence of stable partial-capture equilibria in the intermediate zone. Section 5 gives the exact gain ceilings and the network generalization, where the certificate becomes spectral. Section 6 restates laundering as an exact nonlinear indistinguishability, derives the Kalman rank corollary, and prices the audit sensor by its detection latency. Section 7 proves the separatrix limit: the adaptation rate enters capture risk through an explicit α -free jump map, making update cadence a quantifiable instrument. Section 8 shows the loop parameters are identifiable from contested-stream panels and gives a safe-side certification rule with delta-method inference. Section 9 assembles the regulatory menu; Section 10 records the verification protocol – every theorem, lemma, proposition, and corollary machine-checked in Lean 4, every claim battery-verified; Section 11 states what remains open.

¹A 2026 replication/critique reports that the reward hacking of Wen et al. (2025) disappears in the one setting it reran with a corrected setup (<https://iclr-blogposts.github.io/2026/blog/2026/mislead-lm/>); it is cited here as contested supplementary evidence, and no result below rests on it.

2 The loop model and its classification

2.1 The loop

The reduced stock law of the companion, exact for claim portfolios, gains the content channel:

$$D_{t+1} = (s(\beta_t) + g) D_t + I, \quad s(\beta) = (1 - \lambda_0)(1 - \eta(1 - \beta))^2, \quad (2.1)$$

with grounding rate $\lambda_0 \in (0, 1)$ (the per-period probability that a contested claim is retired by external resolution), responsiveness $\eta \in (0, 1)$, exogenous arrival mass $I > 0$, and content gain $g \geq 0$: each unit of standing disagreement adds g units of fresh contested-claim mass per period. Operator satisfaction is the companion's

$$U(\beta, D) = -v\beta - (1 - c\beta)^2 D, \quad v > 0, c \in (0, 1),$$

and the operator updates by projected gradient ascent at rate α :

$$\beta_{t+1} = \Pi_{[0,1]} \left[\beta_t + \alpha(-v + 2c(1 - c\beta_t)D_t) \right], \quad (2.2)$$

with (2.1) and (2.2) applied *simultaneously*: both updates are computed from (β_t, D_t) . The simultaneous convention matters at the resolution of Section 7, where individual steps remain order one in a singular limit, and it is the convention verified throughout the numerical battery.

The two static characteristics are the population's stationary stock and the operator's best response,

$$D_g^*(\beta) = \frac{I}{1 - s(\beta) - g} \quad (\text{on } s(\beta) + g < 1), \quad \hat{\beta}(D) = \begin{cases} 0, & D \leq \frac{v}{2c}, \\ \frac{1}{c}(1 - \frac{v}{2cD}), & \frac{v}{2c} < D < \frac{v}{2c(1-c)}, \\ 1, & D \geq \frac{v}{2c(1-c)}, \end{cases}$$

both non-decreasing. Loop equilibria are crossings of the composed characteristic $\Phi(\beta) := \hat{\beta}(D_g^*(\beta))$ with the identity, and the crossing pattern is organized by the scalar function

$$G_g(\beta) := -v + 2c(1 - c\beta) \frac{I}{1 - s(\beta) - g}, \quad \Phi(\beta) \geq \beta \iff G_g(\beta) \geq 0. \quad (2.3)$$

2.2 The classification theorem

Assumption 2.1 (Step-size condition (SS_g)). $g < \lambda_0$ and $\alpha < (\lambda_0 - g)/(2c^2 I)$. This is exactly the condition making the map cooperative on the box; it is not by itself a convergence condition. Where convergence is claimed we assume in addition

$$\alpha < \alpha_{\text{mono}}(g) := \frac{(\lambda_0 - g)(s(0) + g)}{2cI[c(s(0) + g) + 2\eta(1 - \lambda_0)]}, \quad (2.4)$$

written (SS_g^+) , which forces $\det J > 0$ uniformly on the box and implies (SS_g) .

Theorem 2.2 (Loop classification). *Under (SS_g) the box $B = [0, 1] \times [I, I/(\lambda_0 - g)]$ is invariant for (2.1)–(2.2) and the map is monotone (cooperative) on it; the larger box $B^* = [0, 1] \times [I, 1/(2\alpha c^2)]$ is invariant, cooperative, and absorbing. If an orbit's states are componentwise comparable at any one time it is monotone thereafter and converges to a fixed point; under (SS_g^+) every orbit converges, and*

the equilibrium set with stabilities is read off the sign pattern of G_g . The step-size hypothesis is not removable: the period-two instance in the sharpness clause of the companion's bistability theorem has $g = 0$ and satisfies (SS_g) strictly (Hofheinz, 2026c). The fixed-point set is read off G_g : the calibrated corner $p_1 = (0, D_g^*(0))$ is a fixed point iff $G_g(0) \leq 0$; the captured corner $p_2 = (1, D_g^*(1))$ iff $G_g(1) \geq 0$; interior fixed points are exactly the roots β of G_g , at $D = D_g^*(\beta)$. At an interior root the characteristic polynomial of the Jacobian satisfies the identity

$$p(1) = -\frac{\alpha D(1-s-g)^2}{I} G'_g(\beta), \quad (2.5)$$

so a transversal descending crossing of G_g ($G'_g < 0$) is a stable fixed point under (SS_g) alone, and a transversal ascending crossing is a saddle provided additionally $2\eta(1-\lambda_0) \leq c(1+s(0)+g)$. The proviso is needed only on the ascending branch, and is not removable there. Corners with strict gradient sign ($G_g(0) < 0$; $G_g(1) > 0$) are locally asymptotically stable.

Proof. *Box.* If $D \leq I/(\lambda_0 - g)$ then $D' \leq (1 - \lambda_0 + g)\frac{I}{\lambda_0 - g} + I = \frac{I}{\lambda_0 - g}$, and $D' \geq I$ always; $\beta' \in [0, 1]$ by projection, so B is invariant. B is not absorbing: at $\beta \equiv 1$ the stock law is exactly $D' = (1 - \lambda_0 + g)D + I$, so $D_0 = \frac{I}{\lambda_0 - g} + \delta$ gives $D_n = \frac{I}{\lambda_0 - g} + (1 - \lambda_0 + g)^n \delta$, strictly above the upper face at every finite n . Take instead $M := 1/(2\alpha c^2)$, the ceiling at which the monotonicity condition $1 - 2\alpha c^2 D \geq 0$ binds; (SS_g) gives $M > I/(\lambda_0 - g)$, whence $(1 - \lambda_0 + g)M + I \leq M$ and $B^* = [0, 1] \times [I, M]$ is invariant, with the monotonicity argument below applying verbatim on it. From any $D_0 \geq 0$ the excess above $I/(\lambda_0 - g)$ contracts by a factor at most $1 - \lambda_0 + g < 1$ per step uniformly in β , so it falls below the strictly positive gap $M - I/(\lambda_0 - g)$ in finitely many steps: B^* is absorbing, and every orbit moreover eventually enters each slack box $[0, 1] \times [I, I/(\lambda_0 - g) + \varepsilon]$, the venue of the convergence argument below.

Monotonicity. On the interior of B the Jacobian of (2.1)–(2.2) is

$$J = \begin{pmatrix} 1 - 2\alpha c^2 D & 2\alpha c(1 - c\beta) \\ s'(\beta) D & s(\beta) + g \end{pmatrix},$$

with $s'(\beta) = 2\eta(1 - \lambda_0)(1 - \eta(1 - \beta)) \geq 0$. All entries are nonnegative: the only one at issue is the first, and on B , $1 - 2\alpha c^2 D \geq 1 - 2\alpha c^2 I/(\lambda_0 - g) \geq 0$ by (SS_g) . Composition with the order-preserving projection $\Pi_{[0,1]}$ preserves monotonicity.

Convergence. If an orbit's states are componentwise comparable at some time, monotonicity propagates the comparison, the orbit is componentwise monotone thereafter, and, confined to a compact invariant box, it converges to a fixed point – order preservation alone. Under (SS_g^+) , suppose instead that no step is comparable, so every increment lies in the open southeast or northwest quadrant. On B the Jacobian entries above satisfy $a \geq a_{\min} = 1 - 2\alpha c^2 I/(\lambda_0 - g)$, $b \leq b_{\max} = 2\alpha c$, $c_J \leq c_{J,\max} = 2\eta(1 - \lambda_0) I/(\lambda_0 - g)$, and $d \geq d_{\min} = s(0) + g$, and (2.4) is exactly $a_{\min} d_{\min} > b_{\max} c_{J,\max}$. The companion's quadrant-exclusion argument (Hofheinz, 2026c) then applies verbatim: decomposing each increment along the legs of its coordinate rectangle, a southeast step can be followed by a northwest step only if $a_{\min} d_{\min} < b_{\max} c_{J,\max}$, and symmetrically for the reverse transition – so the increment quadrant is constant, each coordinate is monotone and bounded, and the orbit converges to a fixed point. Orbits that start above B are covered through the slack boxes, and the admissible slack is exact rather than merely small. Writing $R := I/(\lambda_0 - g)$, the four extrema on $[0, 1] \times [I, R + \varepsilon]$ give strict dominance precisely when $R + \varepsilon < I(s(0) + g)/(\alpha \cdot 2cI[c(s(0) + g) + 2\eta(1 - \lambda_0)])$, that is, when

$$\varepsilon < \varepsilon^* := R \left(\frac{\alpha_{\text{mono}}(g)}{\alpha} - 1 \right) :$$

the box height times the fractional (SS_g^+) margin – the slack one can afford is exactly the slack one has. For any such ε the inflated box is forward invariant, cooperativity there is implied rather than assumed (strict dominance forces $1 - 2\alpha c^2 D > 0$), and the excess of D over R contracts by the factor $1 - \lambda_0 + g$ per step uniformly in β , so entry occurs within $\lceil \log(\varepsilon/(D_0 - R))/\log(1 - \lambda_0 + g) \rceil$ steps. The argument then runs on the inflated box from the entry time on. At the benchmark $\varepsilon^* = 0.134$, a 13.4% inflation of B . Cooperativity alone does not suffice: the period-two instance cited in the statement is the witness. The convergence theory of planar cooperative maps (Smith, 1998) requires an orientation hypothesis that fails here, the clipped projection not being injective.

Fixed points. A fixed point forces $D = D_g^*(\beta)$ from (2.1); then β is fixed under the projected gradient step iff the gradient $-v + 2c(1 - c\beta)D_g^*(\beta) = G_g(\beta)$ vanishes (interior), is ≤ 0 (at $\beta = 0$), or is ≥ 0 (at $\beta = 1$).

Stability. With a, b, c_J, d the entries of J above, $p(1) = (1 - a)(1 - d) - bc_J = 2\alpha cD[c(1 - s - g) - (1 - c\beta)s']$. Differentiating (2.3),

$$G'_g(\beta) = \frac{2cI[-c(1 - s - g) + (1 - c\beta)s']}{(1 - s - g)^2},$$

and substituting $D = I/(1 - s - g)$ at the fixed point yields (2.5). Since $bc_J \geq 0$ the eigenvalues are real, and $\text{tr } J = a + d \leq 1 + (1 - \lambda_0 + g) < 2$, so $p(1) > 0$ places both eigenvalues below 1. For the lower bound, subtracting the two endpoint values of the characteristic polynomial gives the exact identity

$$p(-1) - p(1) = 2(a + d) = 2(1 + s + g - 2\alpha c^2 D),$$

whose right side is positive under (SS_g), since $2\alpha c^2 D < 1$ on the box. At a *descending* crossing $p(1) > 0$ therefore already gives $p(-1) > 0$, with no further hypothesis. At an *ascending* crossing $p(1) < 0$ and a bound is genuinely needed: $p(-1) = (1 + a)(1 + d) - bc_J$, and under (SS_g),

$$bc_J = 2\alpha c(1 - c\beta)s'D \leq \frac{(1 - c\beta)s'}{c} \leq \frac{2\eta(1 - \lambda_0)}{c}, \quad (1 + a)(1 + d) \geq 1 + s(0) + g,$$

so $2\eta(1 - \lambda_0) \leq c(1 + s(0) + g)$ gives $p(-1) > 0$, hence both eigenvalues exceed -1 . (At the benchmark of Section 7 the condition reads $0.98 \leq 1.12$. It is not removable on this branch: the period-two instance of the companion's sharpness clause (Hofheinz, 2026c) satisfies (SS) strictly, yet at its saddle $p(-1) < 0$ and the second eigenvalue drops below -1 – which is exactly the flip that produces the two-cycle.) Thus $G'_g < 0$ at a root implies both eigenvalues in $(-1, 1)$, a stable node, under (SS_g) alone; $G'_g > 0$ implies $p(1) < 0$, one eigenvalue above 1, a saddle, with the second above -1 under the displayed proviso. At a corner with strict gradient sign, continuity gives a neighborhood on which the projected β -step moves weakly toward the corner while D contracts toward D_g^* at rate $s + g < 1$; the corner is locally asymptotically stable. $\square$

Remark 2.3 (Small-gain reading). Theorem 2.2 is the discrete, clipped instance of the characteristic-crossing picture for monotone input/output interconnections (Angeli and Sontag, 2003): D_g^* and $\hat{\beta}$ are the subsystem characteristics, equilibria are their crossings, and multistability is governed by how the composed characteristic crosses the identity (Angeli and Sontag, 2004). What the loop framing adds is that every instrument in Section 9 acts on the characteristics, while the knob only moves the state between them.

3 The aggregation–conservation theorem

Whether stationary measured agreement rewards capture is decided by three structural choices the companion held fixed: which *stock* the population carries (a single claim's mean deviation, or a

portfolio’s L^1 or L^2 mass), which *loss degree* p the metric aggregates, and which *feedback* replenishes the stock (amplitude feedback ζ on a single claim; content gain g on a portfolio). Write $u := 1 - \beta$ and take a single type with $c = 1$ in this section, so the measured weight on standing error is u^p .

Theorem 3.1 (Aggregation–conservation map). *The closed forms below hold unconditionally; the behavior column requires nonzero injection and arrival mass ($\iota \neq 0$ for the single claim, $J > 0$ for the L^1 portfolio, and $I > 0$ for the L^2 portfolio, the last already standing in the model paragraph), since at zero injection the gap vanishes identically and both “iff” clauses fail. The stationary measured gap is, exactly:*

cell	stationary measured gap	behavior in β
single claim, loss $p \in \{1, 2\}$, feedback ζ	$(\iota u/(\eta u - \zeta))^p$	flat iff $\zeta = 0$; rises iff $\zeta > 0$
L^1 portfolio, gain g	$uJ/(\lambda_0 - g + (1 - \lambda_0)\eta u)$	flat iff $g = \lambda_0$; falls iff $g < \lambda_0$
L^2 portfolio, gain g	$u^2I/(1 - (1 - \lambda_0)(1 - \eta u)^2 - g)$	never flat

For the L^2 portfolio, the derivative of the gap in u has the sign of $\lambda_0 + \eta u(1 - \lambda_0) - g$. Hence the gap falls monotonically in β – capture is stationarily rewarded – iff $g \leq \lambda_0$; for $\lambda_0 < g < \lambda_0 + \eta(1 - \lambda_0)$ it attains an interior minimum at

$$u_c = \frac{g - \lambda_0}{\eta(1 - \lambda_0)}, \quad \beta_c = 1 - u_c,$$

beyond which manufactured stock overwhelms the shrinking weight, up to the divergence wall $1 - s(\beta) = g$ (Figure 1).

Proof. Each stationary stock is the geometric sum of its affine law: a single claim’s mean deviation obeys $x' = (1 - \eta u + \zeta)x + \iota$, so $x^* = \iota/(\eta u - \zeta)$ on $\eta u > \zeta$, and the measured gap $(u|x^*|)^p$ is the first row; at $\zeta = 0$ it equals $(|\iota|/\eta)^p$, free of u , and $\frac{d}{du} u/(\eta u - \zeta) = -\zeta/(\eta u - \zeta)^2$ gives the sign claim. An L^1 portfolio obeys $M' = ((1 - \lambda_0)(1 - \eta u) + g)M + J$, so uM^* is the second row, and $\frac{d}{du} u/(\lambda_0 - g + (1 - \lambda_0)\eta u) = (\lambda_0 - g)/(\cdot)^2$: flat exactly at $g = \lambda_0$. For the L^2 portfolio, with $\psi(u) = u^2/(1 - (1 - \lambda_0)(1 - \eta u)^2 - g)$,

$$\psi'(u) \propto 2u \left[(1 - (1 - \lambda_0)(1 - \eta u)^2 - g) - u\eta(1 - \lambda_0)(1 - \eta u) \right] = 2u [\lambda_0 + \eta u(1 - \lambda_0) - g],$$

where the bracket collapses via $(1 - \eta u)[(1 - \eta u) + \eta u] = 1 - \eta u$. The bracket is positive for all u iff $g \leq \lambda_0$; for g in the stated band it changes sign once, at u_c , giving the interior minimum; the stationary stock exists only where $1 - s - g > 0$. $\square$

Remark 3.2 (Where the companion’s knife edge lives). The companion’s conservation point – measured satisfaction flat in β for a single claim without feedback – is the top-left cell at $\zeta = 0$. Theorem 3.1 shows it is one point of a one-parameter conservation *locus* per aggregation ($\zeta = 0$ for single-claim stocks; $g = \lambda_0$ for L^1 portfolios), and that the empirically central case – quadratic loss on a claim portfolio – is never conserved. Below the grounding rate the operator’s stationary landscape genuinely rewards capture, so the usefulness premium v alone carries the trap structure of the companion’s bistable benchmark. Moderately above the grounding rate, the landscape points at a partial-capture interior optimum β_c with finite but inflated stock: an operator who fully internalized the stationary consequences of deference would still choose to stop strictly between calibration and capture. Which feedback structures are dangerous is thus decided by the pair (aggregation, gain), not by the knob.

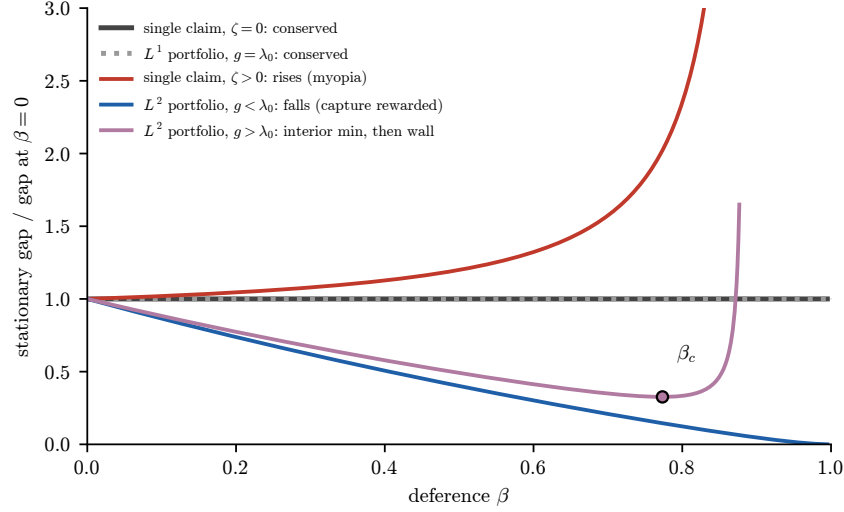


Figure 1: The aggregation–conservation map: stationary measured gap against deference, normalized to its calibrated value. Two conserved loci (flat), the single-claim myopia trap (rising), the quadratic-portfolio capture reward (falling), and the supercritical-gain interior minimum at β_c before the divergence wall.

4 The knob-free certificate

Lemma 4.1 (Policy-free stock bound). *Let $g < \lambda_0$. For any β -trajectory whatsoever – any adaptation rule, any rate, any clipping – $D_t \leq \max(D_0, I/(\lambda_0 - g))$ for all t , and $\limsup_t D_t \leq I/(\lambda_0 - g)$.*

Proof. $s(\beta) + g \leq 1 - \lambda_0 + g < 1$ uniformly in β ; iterate (2.1). $\square$

Definition 4.2 (Sign-consistent rule). An adaptation rule $\beta_{t+1} = R(\beta_t, D_t, \dots)$ is *sign-consistent* if $\beta_{t+1} \leq \beta_t$ whenever $\partial_\beta U(\beta_t, D_t) \leq 0$, and *responsive* if for every $\varepsilon > 0$ there is $\delta(\varepsilon) > 0$ with $\beta_t - \beta_{t+1} \geq \delta(\varepsilon)$ whenever $\partial_\beta U(\beta_t, D_t) \leq -\varepsilon$ and $\beta_t \geq \varepsilon$.

Projected gradient at any rate, fixed-step sign rules, and multiplicative rules are sign-consistent and responsive; heavy-ball momentum becomes so once the gradient is uniformly one-signed, which Corollary 4.4 supplies whenever the margin is strict – only the sharp boundary $m = 0$ is uncovered. The battery (Section 10) exercises all four.

Theorem 4.3 (Knob-free, policy-robust certificate). *Suppose the loop parameters satisfy*

$$m := v(\lambda_0 - g) - 2cI \geq 0. \quad (4.1)$$

Then $G_g(\beta) \leq -vc\beta < 0$ for every $\beta \in (0, 1]$, the calibrated corner is the unique equilibrium of the loop, and every sign-consistent responsive rule, from every initial condition, converges to it. No step-size condition is required.

Proof. For any β and any $D \leq I/(\lambda_0 - g)$,

$$\partial_\beta U = -v + 2c(1 - c\beta)D \leq -v + (1 - c\beta) \frac{2cI}{\lambda_0 - g} \leq -v + (1 - c\beta)v = -vc\beta,$$

using (4.1) in the middle step; taking $D = D_g^*(\beta) \leq I/(\lambda_0 - g)$ gives the bound on G_g , so the only admissible fixed point in Theorem 2.2's catalogue is p_1 . For the dynamics, fix $\varepsilon > 0$. By Lemma 4.1 there is T_0 with $D_t \leq I/(\lambda_0 - g) + \delta_0$ for $t \geq T_0$, where δ_0 is chosen with $2c\delta_0 \leq vc\varepsilon/2$. Then for $t \geq T_0$ and $\beta_t \geq \varepsilon$, $\partial_\beta U \leq -vc\varepsilon + 2c\delta_0 \leq -vc\varepsilon/2 < 0$, so a sign-consistent responsive rule decreases β by at least $\delta(\min\{vc\varepsilon/2, \varepsilon\})$ per step until $\beta_t < \varepsilon$ (the minimum is taken because Definition 4.2 requires the response level to bound both the gradient and β_t , and $\beta_t \geq \varepsilon$ need not give $\beta_t \geq vc\varepsilon/2$), and sign-consistency keeps it there (the gradient is negative whenever $\beta > 0$ and weakly negative at 0). Since ε was arbitrary, $\beta_t \rightarrow 0$, and then $D_t \rightarrow D_g^*(0)$ by the affine recursion. $\square$

Corollary 4.4 (Strict margin: a uniform gradient bound). *If the certificate holds strictly, $m > 0$, then for every $\beta \in [0, 1]$ and every admissible $D \leq I/(\lambda_0 - g)$,*

$$\partial_\beta U \leq -vc\beta - (1 - c\beta) \frac{m}{\lambda_0 - g} \leq -(1 - c) \frac{m}{\lambda_0 - g} < 0.$$

The gradient is thus uniformly one-signed on the whole knob range, including at $\beta = 0$, where the bound of Theorem 4.3 degenerates to zero. At the sharp margin $m = 0$ it does so degenerate; the safe-side rule of Proposition 8.2 never certifies there.

Proof. By (4.1), $2cI/(\lambda_0 - g) = v - m/(\lambda_0 - g)$. Substituting into the chain in the proof of Theorem 4.3,

$$\partial_\beta U \leq -v + (1 - c\beta) \left(v - \frac{m}{\lambda_0 - g} \right) = -vc\beta - (1 - c\beta) \frac{m}{\lambda_0 - g},$$

which is the first inequality. The second uses $1 - c\beta \geq 1 - c > 0$ for $\beta \leq 1$, and $-vc\beta \leq 0$ for $\beta \geq 0$. $\square$

The certificate is *knob-free* – a single inequality among the loop parameters (v, c, I, λ_0, g) , with no reference to β , its trajectory, or the adaptation rule – and *policy-free*: the bound on D in Lemma 4.1 holds for every policy, so no cleverness in the operator's update can defeat it. Its margin m is the natural audit scalar; Section 8 shows m is estimable from panel data with delta-method standard errors.

Theorem 4.5 (Zone structure). *Fix $g < \lambda_0$ and let $F(\beta) := 2c(1 - c\beta)I/(1 - s(\beta) - g)$, so $G_g = -v + F$, and write $F_{\max} = \max_{[0,1]} F$. Then:*

- (A) *If $v \geq F_{\max}$ then $G_g \leq 0$ on $[0, 1]$, and for $v > F_{\max}$ the calibrated corner is the unique equilibrium. Under the certificate (4.1) – which implies $v \geq F_{\max}$, since $F_{\max} \leq 2cI/(\lambda_0 - g)$ – the loop is globally calibrated for every sign-consistent responsive rule (Theorem 4.3); for the gradient loop (2.2) itself, (SS_g^+) together with $v > F_{\max}$ already suffices, by Theorem 2.2.*
- (C) *If $v < F(1) = 2c(1 - c)I/(\lambda_0 - g)$, then $G_g(1) > 0$ and the captured corner is locally asymptotically stable: the companion's capture line, now in loop coordinates.*
- (B) *If $F(1) < v < F_{\max}$, then G_g has exactly one root in $(0, 1)$; it is a transversal descending crossing, $G'_g < 0$ there, and by Theorem 2.2 it is a stable partial-capture equilibrium, unconditionally under (SS_g) – the eigenvalue proviso is automatic on the descending branch: an interior attractor at which deference is pinned strictly between calibration and capture. The strict inequality at the lower boundary is not removable: at $v = F(1)$ an interior equilibrium can fail entirely, as at the admissible parameters $(\eta, \lambda_0, I, v, c, \alpha, g) = (\frac{1}{2}, \frac{3}{20}, \frac{3}{100}, \frac{9}{200}, \frac{9}{10}, 1, \frac{3}{100})$, which*

satisfy every model restriction, strict (SS_g) , the opening condition below, and $v = F(1)$ with $F(\frac{1}{2}) > v$, yet

$$200 G_g(\beta) (303 - 170\beta - 85\beta^2) = 9(1 - \beta)(177 - 85\beta),$$

so $G_g > 0$ on all of $(0, 1)$ while $G_g(1) = 0$: no interior root, hence no partial-capture equilibrium.

Zone (B) is nonempty exactly when F has an interior maximum exceeding $F(1)$; a sufficient condition for F to be decreasing at $\beta = 1$, opening the band, is $c(\lambda_0 - g) > 2\eta(1 - \lambda_0)(1 - c)$. When instead F is increasing on $[0, 1]$ (as at the companion benchmark), $F_{\max} = F(1)$, the band is empty, and the loop dichotomy is the companion's bistable-or-calibrated picture. The three zones in the (g, v) plane, with the computed partial-capture band and the divergence wall $g = \lambda_0$, are shown in Figure 2; both boundary curves blow up as $g \uparrow \lambda_0$, so the certificate price of safety diverges as content gain approaches the grounding rate.

Proof. (A): $v \geq F_{\max}$ gives $G_g \leq 0$ on $[0, 1]$ directly, and for $v > F_{\max}$, $G_g < 0$ everywhere, so Theorem 2.2's catalogue admits no interior fixed point and excludes the captured corner ($G_g(1) < 0$), leaving p_1 ; under (SS_g^+) every orbit of (2.2) converges, hence to p_1 . The all-rule claim under (4.1) is Theorem 4.3, with no transversality needed. (C) is the corner-stability clause of Theorem 2.2. For (B), clear the positive denominator: $q_g(\beta) := G_g(\beta)(1 - s(\beta) - g) = -v(1 - s(\beta) - g) + 2cI(1 - c\beta)$ is a quadratic in β with leading coefficient $v(1 - \lambda_0)\eta^2 > 0$, sharing roots and signs with G_g since $1 - s - g \geq \lambda_0 - g > 0$ on $[0, 1]$. In zone (B), $q_g(1) = (F(1) - v)(\lambda_0 - g) < 0$; if also $q_g(0) \leq 0$, then 0 and 1 both lie between the roots of the positive-leading quadratic, so $q_g \leq 0$ on all of $[0, 1]$, contradicting $F_{\max} > v$. Hence $q_g(0) > 0$, and q_g has exactly one root in $(0, 1)$ – its smaller root – crossed from above: $q'_g < 0$ there, hence $G'_g < 0$, a transversal descending crossing, stable by (2.5). The boundary instance in the statement is checked by direct substitution, the displayed identity clearing denominators exactly. The sign of F' matches the sign of $-c(1 - s - g) + (1 - c\beta)s'$ (the numerator of G'_g), and at $\beta = 1$ this is $-c(\lambda_0 - g) + 2\eta(1 - \lambda_0)(1 - c)$, giving the stated sufficient condition for $F'(1) < 0$. $\square$

Remark 4.6 (Partial capture as an equilibrium phenomenon). The partial-capture equilibria of zone (B) are attractors, not transients or numerical artifacts: intermediate-zone trajectories converge to interior points satisfying the equilibrium condition $G_g = 0$ to machine precision, near $\beta \approx 0.98$ at the benchmark. They are the dynamical counterpart of the stationary interior optimum β_c of Theorem 3.1. For governance they are arguably the most insidious regime: deference is pinned high but not at the corner, the stock is inflated but finite, and no trajectory-level alarm (divergence, corner collapse) ever fires.

5 Gain ceilings and the network certificate

Proposition 5.1 (Exact ceilings). *At full capture the portfolio stock is finite iff $g < \lambda_0$, with $D_g^*(1) = I/(\lambda_0 - g) \uparrow \infty$ as $g \uparrow \lambda_0$ and geometric divergence at rate $1 + g - \lambda_0$ above. In the companion's amplitude model the ceiling is $\zeta_c = \eta e$, the residual evidence coupling at capture; the companion's divergence of the fully-deferent type is the zero-ceiling case $e_L = 0$. The ceiling is a pure loop property: the knob does not appear.*

Proof. At $\beta = 1$, (2.1) reads $D' = (1 - \lambda_0 + g)D + I$; the multiplier crosses 1 at $g = \lambda_0$. The amplitude recursion at capture is $x' = (1 - \eta e + \zeta)x + \iota$, with multiplier crossing 1 at $\zeta = \eta e$. $\square$

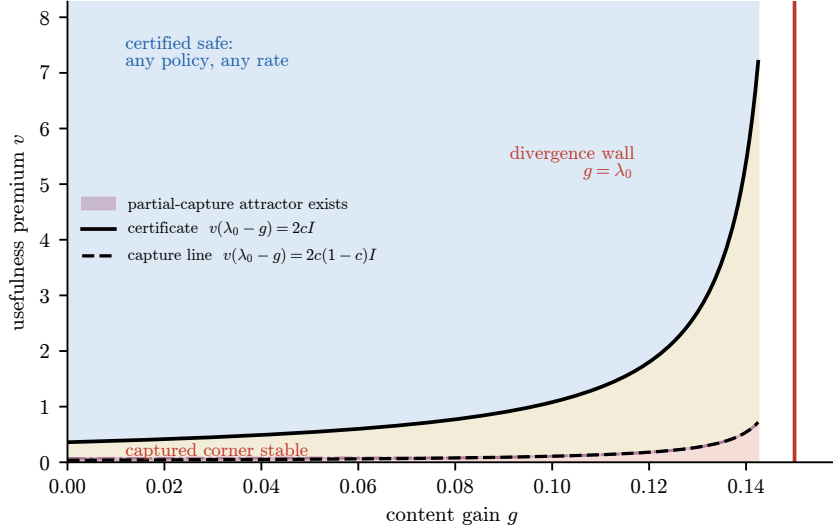


Figure 2: Certificate zones in the loop-parameter plane (g, v) at $\eta = 0.5$, $c = 0.9$, $\lambda_0 = 0.15$, $I = 0.03$. Above the certificate curve: globally calibrated, for any policy at any rate. Below the capture line: the captured corner is stable. The shaded band between carries stable partial-capture equilibria. Both curves diverge at the wall $g = \lambda_0$.

The scalar ceiling generalizes to influence networks, and with it the entire certificate. Let the population comprise K communities with stock vector $D \in \mathbb{R}_{\geq 0}^K$, per-community deference knobs $\beta \in [0, 1]^K$, grounding rates λ_{0k} , responsiveness η_k , and a nonnegative content-coupling matrix $G = (g_{kj})$: disagreement in community j feeds fresh contested claims into community k at gain g_{kj} , the network form of the channel that DeGroot (1974)- and Friedkin and Johnsen (1990)-type influence structures make heterogeneous. The stock law and objective are

$$D_{t+1} = (S(\beta_t) + G)D_t + I, \quad S(\beta) = \text{diag}(s_k(\beta_k)), \quad U = - \sum_k v_k \beta_k - \sum_k (1 - c_k \beta_k)^2 D_k, \quad (5.1)$$

with $s_k(\beta) = (1 - \lambda_{0k})(1 - \eta_k(1 - \beta))^2$ and $\bar{S} := \text{diag}(1 - \lambda_{0k}) \succeq S(\beta)$ entrywise for all β .

Theorem 5.2 (Network certificate). *Suppose $\rho(\bar{S} + G) < 1$ and set $R := (\text{Id} - \bar{S} - G)^{-1}I \in \mathbb{R}_{>0}^K$.*

- (i) Policy-free bound. *For any β -trajectory, $\limsup_t D_t \leq R$ componentwise.*
- (ii) Certificate, static. *If $v_k \geq 2c_k R_k$ for every k , then for all β and all admissible D , $\partial U / \partial \beta_k \leq -2c_k^2 \beta_k R_k < 0$ whenever $\beta_k > 0$.*
- (ii') Certificate, dynamic. *If the margin is uniformly strict $-v_k \geq 2c_k R_k + \mu$ for some $\mu > 0$ and every k – then every per-coordinate sign-consistent responsive rule converges to full calibration $\beta = \mathbf{0}$ from every initial condition. Strictness is not removable at this order: at the sharp margin the coordinate gradient can be marginally positive at small β_k , Definition 4.2 then places no bound on the resulting increase, and communities can stagger their descents so that the coupled stock never enters the exact box $D \leq R$ that the corner argument would need.*
- (iii) Stability ceiling. *At full capture $\beta = \mathbf{1}$ the recursion is linear with nonnegative $M = \bar{S} + G$, and the stock is bounded if and only if the nonnegative series $\sum_{j \geq 0} M^j I$ converges – equivalently,*

by the standard characterization of asymptotic stability for nonnegative matrices, if and only if $\rho(M) < 1$. Boundedness in particular forces $M^n I \rightarrow 0$. The scalar case $K = 1$ recovers $R = I/(\lambda_0 - g)$, the certificate (4.1), and the ceiling $g < \lambda_0$ exactly.

- (iv) Decentralized sufficient condition. $\sum_j g_{kj} < \lambda_{0k}$ for every k implies $\rho(\bar{S} + G) < 1$: per-community gain below per-community grounding certifies the spectral condition without computing a spectrum.

Proof. (i) Since $S(\beta) \preceq \bar{S}$ entrywise, $D \geq 0$, and $G \geq 0$, induction gives $D_t \leq (\bar{S} + G)^t D_0 + \sum_{j < t} (\bar{S} + G)^j I$; the Neumann series converges to R when $\rho(\bar{S} + G) < 1$, all terms nonnegative. (ii) $\partial U / \partial \beta_k = -v_k + 2c_k(1 - c_k \beta_k) D_k \leq -v_k + 2c_k(1 - c_k \beta_k) R_k \leq -2c_k^2 \beta_k R_k$ using $v_k \geq 2c_k R_k$. (ii') By (i), for any $\varepsilon > 0$ each coordinate satisfies $D_{t,k} \leq R_k + \varepsilon$ from some time T_k on, and since K is finite the common time $T = \max_k T_k$ exists. Taking $\varepsilon = \mu/8$ and using $1 - c_k \beta_k \leq 1$, for $t \geq T$

$$\partial U / \partial \beta_k \leq -v_k + 2c_k(R_k + \varepsilon) \leq -\mu + \frac{\mu}{2} = -\frac{\mu}{2} < 0$$

uniformly in β , including at $\beta_k = 0$. Sign-consistency therefore forbids any coordinate from increasing after T , so each $\beta_{t,k}$ is non-increasing and bounded below; responsiveness applied at level $\min\{\mu/2, b\}$ drives it below any $b > 0$. Note that no entry into the exact box $D \leq R$ is required, and hence no synchronization across communities: it is uniform negativity of the gradient, not membership of the box, that closes the argument. (iii) At $\beta = \mathbf{1}$, $S = \bar{S}$ and $D_t = M^t D_0 + \sum_{j < t} M^j I$ with every term nonnegative. The partial sums of $\sum_j M^j I$ are then nondecreasing and dominated by D_t , so a bounded stock makes the series summable and its terms vanish. Conversely, if the series converges then, since $I \succ 0$ gives $I \geq \delta \mathbf{1}$ for some $\delta > 0$, the nonnegative series $\sum_j M^j \mathbf{1}$ converges as well; hence $M^j \mathbf{1} \rightarrow 0$, so $M^t D_0 \leq \|D_0\|_\infty M^t \mathbf{1} \rightarrow 0$ and the stock is bounded. Both directions are elementary and use no spectral theory; the equivalence with $\rho(M) < 1$ is the standard nonnegative-matrix characterization, quoted for interpretation only – every applied condition is stated through the row-sum bound of (iv) or the induced operator norm. (iv) $\rho(M) \leq \max_k \sum_j M_{kj} = \max_k (1 - \lambda_{0k} + \sum_j g_{kj}) < 1$. $\square$

The conceptual content of (iii) is that *the spectral gap of the dissipation-plus-coupling operator replaces the scalar margin $\lambda_0 - g$* : in a connected influence structure, one community's excess gain can be compensated by another's grounding only up to the spectral radius $\rho(\bar{S} + G)$ – spectral compensation, not a defined economic transfer – and the wisdom-of-crowds intuition that diversified structures dampen amplification (Golub and Jackson, 2010) becomes the precise statement that what must stay below one is $\rho(\bar{S} + G)$, not any single entry.

6 Laundering as an observability failure

The companion's laundering theorem says that at capture the satisfaction weight on the most-harmed group vanishes identically while that group's error diverges. In loop language this is a statement about the measurement channel, and it is stronger than its linearization. Work on the captured face of the companion's two-type model: type H retains residual evidence coupling $\kappa \in (0, 1)$, so its standing deviation contracts, $x'_H = a_H x_H + \iota_H$ with $a_H = 1 - \eta_H \kappa + \zeta < 1$; type L 's coupling has vanished, $x'_L = a_L x_L + \iota_L$ with $a_L = 1 + \zeta > 1$; and the satisfaction observable weights each type's error by its residual independence squared,

$$y = -\pi_H \kappa^2 x_H^2 - \pi_L \cdot 0^2 \cdot x_L^2 = -\pi_H \kappa^2 x_H^2.$$

Theorem 6.1 (Exact nonlinear indistinguishability). *On the captured face the state map is decoupled, $T(x_H, x_L) = (T_H(x_H), T_L(x_L))$, and the satisfaction output depends on x_H alone. Hence for every t , $y_t = h(T_H^t(x_{H,0}))$: the entire satisfaction output sequence is a function of the healthy coordinate's initial condition, exactly, globally, and without linearization. Any two face states agreeing in x_H are output-indistinguishable for all time, while their x_L coordinates diverge geometrically at rate $1 + \zeta$. Capture is an unobservable instability of the satisfaction channel in the full nonlinear sense.*

Proof. Immediate from the two displayed structural facts: decoupled dynamics propagate x_H autonomously, and y is a function of x_H only, so $x_{L,0}$ never enters any y_t . $\square$

Corollary 6.2 (Kalman rank, audit sensors, and the calibrated contrast). *Linearizing at the face equilibrium $x_H^* = \iota_H/(\eta_H\kappa - \zeta)$: $A = \text{diag}(a_H, a_L)$ and $C_1 = (-2\pi_H\kappa^2x_H^*, 0)$. For $x_H^* \neq 0$ – equivalently $\iota_H \neq 0$, under the standing positive denominator – rank $\mathcal{O}(C_1, A) = 1$ and the unstable eigenvector e_L spans the unobservable subspace; at $\iota_H = 0$ the sensitivity vanishes, $C_1 = 0$, and the rank is 0: the satisfaction channel is then blind to both modes, not merely the diverging one. Augmenting with an ε -weighted unweighted-error audit sensor $y_2 = -\varepsilon(\pi_Hx_H^2 + \pi_Lx_L^2)$ gives rank $\mathcal{O} = 2$ for every $\varepsilon > 0$ at every face point with $x_H^* \neq 0$ and $x_L \neq 0$. At $x_H^* = 0$ the stacked rank is 1, yet the sole unobservable mode is the contracting e_H -mode with multiplier $a_H < 1$, so the augmented pair is detectable without being observable: the audit still sees every unstable direction, which is what the alarm needs. Detectability is binary in ε . At the calibrated equilibrium, distinct effective contraction rates together with nonzero linearized sensitivities for both types make the satisfaction sensor alone observable; laundering is specifically a capture-state pathology, not a generic deficiency of satisfaction measurement. (With equal rates the calibrated pair has a repeated eigenvalue and a single scalar output cannot separate the modes – rank one when the sensor is nonzero, zero when it is not: a degeneracy of the test configuration, not of the theorem.)*

Theorem 6.3 (Exact detection latency). *Along the captured face with $\zeta > 0$, nonzero sign-aligned injection, and $x_L(t) = (x_{L,0} - \bar{x})(1 + \zeta)^t + \bar{x}$, where $\bar{x} = -\iota_L/\zeta$, set*

$$r := 1 + \zeta, \quad a := |x_{L,0} - \bar{x}|, \quad b := |\bar{x}|, \quad R_\varepsilon := \sqrt{\frac{\theta}{\varepsilon\pi_L}}.$$

For every $\varepsilon, \pi_L, \theta > 0$ satisfying $R_\varepsilon \geq \max\{b, |x_{L,0}|\}$, the first integer t with $\varepsilon\pi_Lx_L(t)^2 \geq \theta$ is exactly

$$T_{\text{det}}(\varepsilon) = \left\lceil \frac{\ln((R_\varepsilon + b)/a)}{\ln r} \right\rceil.$$

Writing

$$L_\varepsilon := \frac{\ln(\theta/(\varepsilon\pi_L a^2))}{2 \ln r}, \quad \delta_\varepsilon := \frac{\ln(1 + b/R_\varepsilon)}{\ln r},$$

the integer remainder obeys the explicit bounds

$$\delta_\varepsilon \leq T_{\text{det}}(\varepsilon) - L_\varepsilon < 1 + \delta_\varepsilon, \quad 0 \leq \delta_\varepsilon \leq \frac{\ln 2}{\ln(1 + \zeta)}.$$

Consequently, as $\varepsilon \rightarrow 0^+$ at fixed $(\pi_L, \theta, \zeta, \iota_L, x_{L,0})$, latency is linear in $\ln(1/\varepsilon)$ with slope $1/(2 \ln(1 + \zeta))$ and a uniformly bounded remainder. Because the sensor reads x_L^2 , it identifies $|x_L|$; the harm is symmetric in the sign of the deviation, so this parity blindness is immaterial.

Proof. Sign alignment gives $|x_L(t)| = ar^t - b$ and $a = |x_{L,0}| + b$. The alarm condition is therefore $R_\varepsilon \leq ar^t - b$, equivalently $\ln((R_\varepsilon + b)/a)/\ln r \leq t$. The stated range makes this real crossing time nonnegative, so its first integer solution is its ceiling. The identity

$$\frac{\ln((R_\varepsilon + b)/a)}{\ln r} = L_\varepsilon + \delta_\varepsilon$$

and $q \leq [q] < q + 1$ give the remainder bracket. Finally $b/R_\varepsilon \in [0, 1]$ on the stated range, which gives the displayed $\ln 2/\ln r$ bound. $\square$

Numerical Result 6.4. In the battery the fitted latency slope is 41 against the predicted 42 with $R^2 > 0.999$, and the asymptotic growth rate of the sensor gap matches $2 \ln(1 + \zeta)$ to 0.4%.

Remark 6.5 (Sensor mandate). Detectability is binary in ε ; latency is logarithmic. A regulator therefore chooses ε from a latency target rather than a detectability one:

$$\varepsilon \geq \frac{\theta}{\pi_L(a(1 + \zeta)^{T_{\text{target}}} - b)^2}, \quad a(1 + \zeta)^{T_{\text{target}}} > b.$$

This condition is necessary and sufficient for an alarm by the target time in the sign-aligned captured-face model. It is the formal content of an audit-sensor mandate, and Theorem 6.1 is the reason it cannot be replaced by any amount of satisfaction telemetry: the laundered channel carries literally zero bits about the diverging coordinate. In the explicit finite deterministic-channel model, for any finite prior over low-coordinate states and any finite quantization of the satisfaction output, mutual information equals output Shannon entropy and is exactly zero; this consequence of Theorem 6.1 is machine-checked in `PaperIII/DeterministicInformation.lean`.

7 Rate separation: the separatrix limit

The companion's bistable benchmark – $\eta = 0.5$, $c = 0.9$, $\lambda_0 = 0.02$, $I = 0.02$, $v = 0.10$, $g = 0$, saddle at $\beta^\dagger = 0.9754$ – separates calibration from capture by a basin boundary in (β, D) . How high that boundary sits in D , from a transient initial condition (β_0, D_0) with excess stock, depends on the adaptation rate: a fast operator ratchets β through the transient before the stock relaxes; a slow one averages over the stationary landscape, where $G_g < 0$ below the saddle. The limit is not, as one might first guess, a continuum flow.

Theorem 7.1 (Jump-map transit limit). *Assume $g < \lambda_0$ and rescale the stock excess as $\tilde{E} := \alpha(D - D_g^*(\beta))$. One simultaneous step of (2.1)–(2.2) is then, exactly and for every $\alpha \neq 0$,*

$$\beta' = \Pi_{[0,1]}(\beta + A(\beta)\tilde{E} + \alpha G_g(\beta)), \quad \tilde{E}' = \sigma(\beta)\tilde{E} + \alpha(D_g^*(\beta) - D_g^*(\beta')),$$

with $A(\beta) = 2c(1 - c\beta)$ and $\sigma(\beta) = s(\beta) + g$. Consequently, from fixed rescaled data $x_0 = (\beta_0, \tilde{E}_0)$ with $\beta_0 \in [0, 1]$ – initial stock $D_0 = D_g^*(\beta_0) + \tilde{E}_0/\alpha$ – the trajectory converges as $\alpha \rightarrow 0^+$, at every fixed finite horizon, to the orbit of the α -free projected jump map

$$(\beta, \tilde{E}) \mapsto (\Pi_{[0,1]}(\beta + A(\beta)\tilde{E}), \sigma(\beta)\tilde{E}), \quad (7.1)$$

whose individual β -steps remain order one in the limit. On transits all of whose raw policy inputs lie in $[0, 1]$ the projection is inactive and (7.1) is the unprojected affine jump map; the projection is not removable in general, since a raw jump past one makes the two maps differ.

More strongly, write $x_t^\alpha = (\beta_t^\alpha, \tilde{E}_t^\alpha)$ for the displayed exact rescaled recursion and x_t^0 for the projected jump orbit from the same x_0 . There is a finite constant C , depending only on the fixed primitives and x_0 , such that for every $\alpha > 0$ and $t \in \mathbb{N}$,

$$\|x_t^\alpha - x_t^0\|_\infty \leq C \alpha t.$$

If $\tilde{E}_0 > 0$, set $q := 1 - \lambda_0 + g \in (0, 1)$ and, for $0 < \epsilon \leq \tilde{E}_0$,

$$H_\epsilon := \left\lceil \frac{\log(\tilde{E}_0/\epsilon)}{\log(1/q)} \right\rceil, \quad L_\epsilon := 1 + \frac{\log(\tilde{E}_0/\epsilon)}{\log(1/q)}.$$

Then $|\tilde{E}_{H_\epsilon}^0| \leq \epsilon$ and, uniformly over the named projected transit,

$$\max_{0 \leq t \leq H_\epsilon} \|x_t^\alpha - x_t^0\|_\infty \leq C \alpha L_\epsilon.$$

Moreover, the explicit joint range

$$C \alpha L_{\epsilon/2} < \epsilon/2$$

guarantees $|\tilde{E}_{H_{\epsilon/2}}^\alpha| < \epsilon$. Since

$$L_\epsilon = 1 + \frac{\log(\tilde{E}_0/\epsilon)}{\log(1/q)},$$

the displayed inequality is the precise growing-horizon bound. Along every joint family with $\epsilon \rightarrow 0^+$ and $\alpha \log(1/\epsilon) \rightarrow 0$, its right-hand side tends to zero uniformly over $0 \leq t \leq H_\epsilon$.

Proof. Substituting $D = D_g^*(\beta) + \tilde{E}/\alpha$ into (2.1)–(2.2) and using $\sigma D_g^* + I = D_g^*$ gives the displayed pair with no expansion: the conjugacy is exact at every $\alpha \neq 0$.

For the uniform estimate, put $d := \lambda_0 - g$, $B := I/d$, $w := 1 + 2c/d$, $\Xi := 2c^2 + w 2\eta(1 - \lambda_0)$, and $F := v + 2cB + wB$. The characteristic lies in $[0, B]$, $|G_g| \leq v + 2cB$, $|A| \leq 2c$, A is $2c^2$ -Lipschitz, σ is $2\eta(1 - \lambda_0)$ -Lipschitz, and the policy projection is 1-Lipschitz. Hence, with

$$b_t := |\beta_t^\alpha - \beta_t^0|, \quad e_t := |\tilde{E}_t^\alpha - \tilde{E}_t^0|,$$

the two exact recursions give

$$\begin{aligned} b_{t+1} &\leq b_t + 2c e_t + 2c^2 |\tilde{E}_t^0| b_t + \alpha(v + 2cB), \\ e_{t+1} &\leq q e_t + 2\eta(1 - \lambda_0) |\tilde{E}_t^0| b_t + \alpha B. \end{aligned}$$

The projected excess satisfies $|\tilde{E}_t^0| \leq |\tilde{E}_0| q^t$, while $2c + wq \leq w$. Therefore the weighted error $R_t := b_t + w e_t$ obeys

$$R_{t+1} \leq (1 + \Xi |\tilde{E}_0| q^t) R_t + \alpha F, \quad R_0 = 0.$$

Iteration and $1 + u \leq e^u$ yield

$$R_t \leq \alpha F t \prod_{k < t} (1 + \Xi |\tilde{E}_0| q^k) \leq \alpha F t \exp\left(\frac{\Xi |\tilde{E}_0|}{d}\right).$$

Since $w \geq 1$, both coordinate errors are at most R_t ; take $C := F \exp(\Xi |\tilde{E}_0|/d)$. This proves the all-horizon bound and hence the fixed-horizon limit.

Finally, $|\tilde{E}_t^0| \leq \tilde{E}_0 q^t$ and the definition of H_ϵ give $|\tilde{E}_{H_\epsilon}^0| \leq \epsilon$ and $H_\epsilon < L_\epsilon$. Applying the all-horizon estimate at every $t \leq H_\epsilon$ proves the uniform logarithmic-transit bound. At $H_{\epsilon/2}$, the triangle inequality gives $|\tilde{E}_{H_{\epsilon/2}}^\alpha| \leq \epsilon/2 + C \alpha L_{\epsilon/2} < \epsilon$ on the displayed joint range. $\square$

Remark 7.2 (Separatrix scale). Write

$$\kappa^*(\beta_0) := \sup \left\{ \tilde{E}_0 : \sup_t \beta_t < \beta^\dagger \text{ under (7.1)} \right\}.$$

Under (SS_g^+) every orbit of the full loop converges and limits are monotone in the initial stock, so at each β_0 the captured initial stocks form an up-set and $D_{\text{sep}}(\alpha; \beta_0)$ is well defined as its infimum. Proposition 7.6 proves the limit $\alpha(D_{\text{sep}} - D_g^*(\beta_0)) \rightarrow \kappa^*(\beta_0)$; the battery measures the rescaled threshold descending onto κ^* across six rates (Section 10).

Lemma 7.3 (Characteristic capture). *In a bistable configuration of the loop – fixed points $p_1 < p_2 < p_3$ with saddle policy $\beta^\dagger$ – every state (β, D) with $\beta > \beta^\dagger$ and $D \geq D_g^*(\beta)$ lies in the basin of the captured corner, for every α satisfying (SS_g) .*

Proof. From $(\beta, D_g^*(\beta))$ the first step is componentwise comparable: $G_g(\beta) > 0$ above the saddle moves the policy weakly up, and the stock is exactly stationary on the characteristic. By the comparable-step principle of Theorem 2.2 the orbit is monotone and converges to a fixed point with policy at least $\beta > \beta^\dagger$, which the catalogue forces to be p_3 . For D above the characteristic, monotonicity of fate is not by itself enough: the statement places no upper bound on D , while the policy coordinate is not monotone in its own input above the monotonicity ceiling. Split instead on the stationary ceiling $D_g^*(1) = I/(\lambda_0 - g)$. At or below it the state lies in B and the comparison above applies verbatim. Above it there are two cases. If the orbit's stock never returns to the ceiling, then $\partial_\beta U > G_g(1) > 0$ at every state it visits, since $1 - c\beta \geq 1 - c$ and $D > D_g^*(1)$; the policy therefore rises by at least $\alpha G_g(1)$ per step until it clips at 1, after which the stock contracts to $D_g^*(1)$. If it first returns at some time T , the policy is nondecreasing up to T by the same gradient bound, so the entry state dominates $(\beta_{T-1}, D_g^*(\beta_{T-1}))$ componentwise with $\beta_{T-1} \geq \beta > \beta^\dagger$, and the characteristic-orbit argument applies to it inside B . Forward invariance of the region above the characteristic is not an available substitute: at an unclipped state exactly on the characteristic the stock is stationary while the policy rises, so the next state lies strictly below the new characteristic. $\square$

Proposition 7.4 (Burn decomposition: the scale is a Riemann sum of its integral). *Let $w(\beta) := (1 - \sigma(\beta))/A(\beta)$ and consider jump orbits (7.1) from β_0 with $\tilde{E}_0 \in [0, 1/(2c^2))$.*

- (i) *Along any such orbit that never clips, the burn telescopes exactly: $\tilde{E}_0 = \sum_{t \geq 0} w(\beta_t) (\beta_{t+1} - \beta_t)$.*
- (ii) *The policy limit $\beta_\infty(\tilde{E}_0)$ is continuous and non-decreasing on $[0, 1/(2c^2))$. The excesses whose orbits never clip form an initial segment; β_∞ is strictly increasing there and identically 1 above it. Strictness is not global: at the benchmark from $\beta_0 = 0$, both $\tilde{E}_0 = 3/5$ and $\tilde{E}_0 = 61/100$ lie in $[0, 50/81) = [0, 1/(2c^2))$ and give raw first policy inputs $27/25$ and $549/500$, so both clip on the first step and share the limit one; at $\beta_0 = 0.4$ the plateau covers 61% of the stated interval. The plateau nevertheless sits strictly above the critical excess. If the transit brackets the saddle – $\beta_0 < \beta^\dagger < \beta_\infty(\tilde{E}_0)$ for some admissible $\tilde{E}_0$ – then $\kappa^*(\beta_0)$ is the unique root of $\beta_\infty = \beta^\dagger$; since $\beta_\infty(\kappa^*) = \beta^\dagger < 1$, the critical orbit is unclipped with $\beta_t \uparrow \beta^\dagger$ and κ^* lies strictly inside the strictly increasing segment, so the separation is two-sided – $\beta_\infty < \beta^\dagger$ below κ^* and $\beta_\infty > \beta^\dagger$ above it. Finally*

$$\kappa^*(\beta_0) = \sum_{t \geq 0} w(\beta_t) (\beta_{t+1} - \beta_t) :$$

the exact left-endpoint Riemann sum of $K(\beta_0) := \int_{\beta_0}^{\beta^\dagger} w(b) db$ over the critical orbit's own partition of $[\beta_0, \beta^\dagger]$.

(iii) The jumps $h_t = A(\beta_t)\tilde{E}_t$ along the critical orbit are strictly decreasing, and

$$|\kappa^*(\beta_0) - K(\beta_0)| \leq h_0 V(w) = A(\beta_0) \kappa^*(\beta_0) V(w), \quad V(w) := \int_{\beta_0}^{\beta^\dagger} |w'(b)| db,$$

with $K \leq \kappa^*$ when w decreases on $[\beta_0, \beta^\dagger]$, as at the benchmark. Consequently $\kappa^*(\beta_0)/K(\beta_0) \rightarrow 1$ as $\beta_0 \rightarrow \beta^{\dagger-}$; once $A(\beta_0)V(w) < 1$, moreover, $|\kappa^*/K - 1| \leq A(\beta_0)V/(1 - A(\beta_0)V)$.

Proof. (i) $\tilde{E}_t - \tilde{E}_{t+1} = (1 - \sigma(\beta_t))\tilde{E}_t = w(\beta_t)A(\beta_t)\tilde{E}_t = w(\beta_t)(\beta_{t+1} - \beta_t)$, and $\tilde{E}_t \leq \tilde{E}_0 \bar{\sigma}^t \rightarrow 0$ with $\bar{\sigma} := 1 - \lambda_0 + g < 1$; telescope. (ii) On $\{0 \leq \tilde{E} < 1/(2c^2)\}$ the jump map is cooperative – $\partial\beta'/\partial\beta = 1 - 2c^2\tilde{E} > 0$ and the other three partials are nonnegative – β_t is nondecreasing, and the tails $\beta_\infty - \beta_T \leq A(0)\tilde{E}_0\bar{\sigma}^T/(1 - \bar{\sigma})$ are uniform on compacts, so β_∞ is a uniform limit of continuous maps, and non-decreasing in $\tilde{E}_0$ because the projection is order-preserving (Hofheinz, 2026c). Where the larger orbit never clips the smaller does not either, both are then raw orbits, and their ordered policy gaps b_t obey $b_{t+1} \geq b_t(1 - 2c^2\tilde{E}_t')$ with $b_1 = A(\beta_0)(\tilde{E}'_0 - \tilde{E}_0) > 0$ and $\sum_t \tilde{E}'_t < \infty$, so $b_\infty \geq b_1 \prod_{t \geq 1} (1 - 2c^2\tilde{E}'_t) > 0$: strict there. Above that segment the raw input stays at least one, since $A > 0$ and $\tilde{E} \geq 0$, so the face is absorbing and $\beta_\infty \equiv 1$; the displayed plateau instance is checked by direct substitution. Because $\beta^\dagger < 1$, the level $\beta^\dagger$ is attained inside the strictly increasing segment: the intermediate value theorem gives the root, strictness gives its uniqueness and the two-sided separation, and the critical orbit never clips because its policy increases to $\beta^\dagger < 1$. (iii) $h_{t+1}/h_t = \sigma(\beta_t)A(\beta_{t+1})/A(\beta_t) < 1$. Subtracting the integral from the sum,

$$\kappa^* - K = \sum_t \int_{\beta_t}^{\beta_{t+1}} (w(\beta_t) - w(b)) db,$$

each term is at most h_t times the variation of w on $[\beta_t, \beta_{t+1}]$, and $\sum_t h_t V_t \leq h_0 V$; when w decreases every term is nonnegative. The ratio bound is arithmetic. $\square$

Lemma 7.5 (Sub-saddle trap, uniformly in the adaptation rate). *Fix a bistable configuration and $\rho \in (0, \beta^\dagger)$, and set*

$$m := - \max_{[0, \beta^\dagger - \rho/2]} G_g > 0, \quad L := \max_{[0, \beta^\dagger - \rho/2]} (D_g^*)', \quad \bar{G} := \max_{[0, 1]} |G_g|, \quad \varepsilon^* := \frac{\rho(1 - \bar{\sigma})}{16c}.$$

There is an explicit $\alpha_0 > 0$, depending only on these constants, such that for every $\alpha \leq \alpha_0$: every full-loop state (β, D) with $\beta \leq \beta^\dagger - \rho$ and $\alpha(D - D_g^(\beta)) \leq \varepsilon^*$ converges to the calibrated corner.*

Proof. Write $E_t := D_t - D_g^*(\beta_t)$, $E_t^+ := \max(E_t, 0)$, $d := 1 - \bar{\sigma} = \lambda_0 - g$, and $q := \bar{\sigma}$. Bound the moving-characteristic correction *directionally*. Absolute policy movement is not controlled by $\bar{G}$ where $E_t < 0$, the transient gradient there being more negative than $G_g(\beta_t)$, and the hypothesis is only an upper bound on αE_0 ; but if the policy moves up, D_g^* is increasing and the correction is nonpositive, while if it moves down, $D_t \geq 0$ gives $\partial_\beta U \geq -v$, so the projection lowers the policy by at most αv . With L any positive Lipschitz bound for D_g^* – the reachable-interval maximum above, or the global $2\eta(1 - \lambda_0)I/(\lambda_0 - g)^2$ – this gives the one-sided recurrence

$$E_{t+1}^+ \leq qE_t^+ + \alpha Lv.$$

valid on the whole initial-state domain. Take

$$\alpha_0 := \min \left\{ \frac{1 - \bar{\sigma}}{4c^2 I}, \frac{m(1 - \bar{\sigma})}{16cLv} \right\},$$

which implies (SS_g) strictly and makes the error floor at most $m/(16c)$. The coupled induction used by the formal proof gives, for every t before any putative escape from the sub-saddle interval,

$$E_t^+ \leq q^t E_0^+ + \frac{\alpha L v}{d}, \quad \beta_t \leq \beta_0 + \frac{8c \alpha E_0^+}{3d} (1 - q^t).$$

The initial allowance gives

$$\frac{8c \alpha E_0^+}{3d} \leq \frac{\rho}{6},$$

so $\beta_t \leq \beta^\dagger - \rho + \rho/6 < \beta^\dagger - \rho/2$ for every t . This closes the induction globally; no logarithmic cutoff and no additional ceiling on α are needed. Since $q^t E_0^+ \rightarrow 0$ and the error floor is at most $m/(16c)$, eventually $E_t^+ \leq m/(8c)$. On the invariant sub-saddle interval, $G_g(\beta_t) \leq -m$ and $A(\beta_t) \leq 2c$, hence $\partial_\beta U(\beta_t, D_t) \leq -m + 2cE_t^+ \leq -m/2$. The policy therefore reaches 0 in finitely many further steps and remains there; the stock recursion then contracts to $D_g^*(0)$, so the orbit converges to p_1 . $\square$

Proposition 7.6 (The separatrix limit). *In a bistable configuration under (SS_g^+) , at every β_0 with a saddle-bracketing transit,*

$$\alpha (D_{\text{sep}}(\alpha; \beta_0) - D_g^*(\beta_0)) \longrightarrow \kappa^*(\beta_0) \quad (\alpha \rightarrow 0^+).$$

Proof. Upper half. Fix $\delta > 0$. The jump orbit from $\tilde{E}_0 = \kappa^* + \delta$ has policy limit above $\beta^\dagger$ by Proposition 7.4(ii), so some finite t^* has $\beta_{t^*} > \beta^\dagger$, and $\tilde{E}_{t^*} > 0$. By Theorem 7.1 at horizon t^* , for all small α the rescaled full-loop orbit lands within $\frac{1}{2}(\beta_{t^*} - \beta^\dagger)$ and $\frac{1}{2}\tilde{E}_{t^*}$ of the jump state: a state with policy above $\beta^\dagger$ and stock above the characteristic, captured by Lemma 7.3. Monotonicity of fate gives $D_{\text{sep}} \leq D_g^*(\beta_0) + (\kappa^* + \delta)/\alpha$.

Lower half. Fix $\delta > 0$ and let $2\rho := \beta^\dagger - \beta_\infty(\kappa^* - \delta) > 0$, again by Proposition 7.4(ii). Choose T with jump excess $\tilde{E}_T \leq \varepsilon^*(\rho)/2$; by Theorem 7.1 at horizon T , for all small α the rescaled full-loop orbit lands with policy below $\beta^\dagger - \rho$ and rescaled excess below $\varepsilon^*(\rho)$, and Lemma 7.5 calibrates it. Monotonicity of fate gives $D_{\text{sep}} \geq D_g^*(\beta_0) + (\kappa^* - \delta)/\alpha$. Together, $\alpha (D_{\text{sep}} - D_g^*(\beta_0)) \in [\kappa^* - \delta, \kappa^* + \delta]$ for all small α . $\square$

Remark 7.7 (Why K is an approximation, not a bound). The jump map evaluates the burn rate $(1 - \sigma)/A$ at the *left endpoint* of each order-one jump. When the integrand is decreasing in β – as at the benchmark – left-endpoint sampling overestimates the burn, pushing κ^* slightly above K ; the orientation reverses when the integrand increases. The discrepancy is a Riemann-sum error at the jump scale, not a one-sided inequality, and vanishes with the jump size. The discrete update convention matters at the same order: the simultaneous-update map (2.1)–(2.2) yields (7.1) with σ evaluated at the pre-jump β ; a sequential convention shifts σ to the post-jump β and changes κ^* at order one. The convention is therefore part of the statement.

At the benchmark with $\beta_0 = 0.4$: $\kappa^* = 0.2384$ and $K = 0.2366$ (within 0.8%); the measured products $\alpha (D_{\text{sep}} - D^*(\beta_0))$ are

α	0.6	0.3	0.15	0.075	0.0375	0.01875
$\alpha (D_{\text{sep}} - D^*)$	0.525	0.396	0.326	0.287	0.265	0.253

monotone in α with Richardson extrapolation 0.2405, within 0.9% of κ^* (Figure 3); near the saddle ($\beta_0 = 0.9$), where jumps are small, $\kappa^* = 0.0213$ against $K = 0.0211$, confirming the continuum regime.

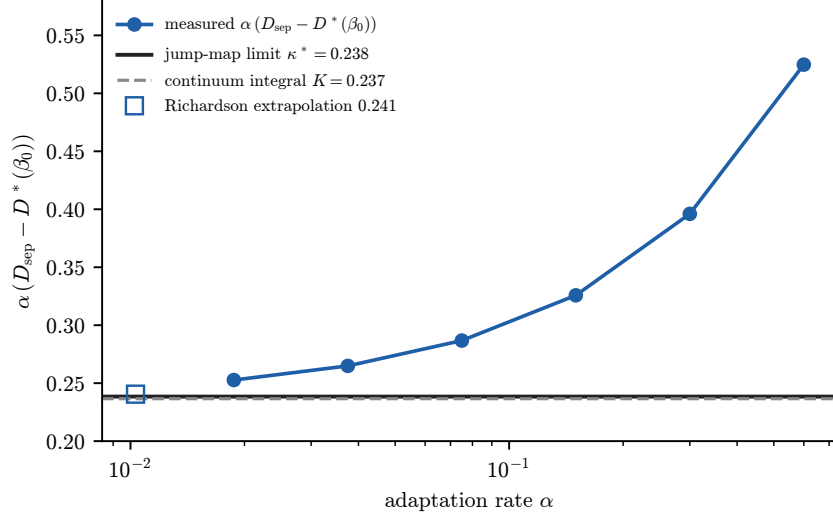


Figure 3: The separatrix limit at the benchmark, $\beta_0 = 0.4$: measured $\alpha(D_{\text{sep}} - D^*(\beta_0))$ descends monotonically onto the jump-map separatrix scale κ^* as $\alpha \rightarrow 0$; the continuum integral K lies within 0.8%, and the Richardson extrapolation of the two smallest rates lands within 0.9% of κ^* .

Remark 7.8 (Cadence as a regulable). The separatrix limit (Proposition 7.6) prices the adaptation rate: from a transient shock of excess stock E_0 , capture requires $\alpha \gtrsim \kappa^*(\beta_0)/E_0$, so capping preference-update cadence relative to belief-mixing time buys separatrix height in inverse proportion in the small- α limit. This is the deterministic skeleton of the two-timescale principle in stochastic approximation (Borkar, 1997, 2008): there, separating timescales is an algorithm-design device; here, the timescale ratio is itself the regulated quantity, and the jump map is the object that prices it.

8 Identification and the safe-side rule

The certificate is only as useful as its parameters are estimable. The task is system identification in the standard sense – recovering a dynamic system’s parameters from input–output records under sufficient excitation (Ljung, 1999) – with two features that decide the design: the excitation is the deference path, which an operator at an attractor holds still, and the estimand of record is not the fitted model but a one-sided inequality on it, so the relevant error criterion is the sign of a margin rather than a norm. Consider a contested-stream panel: per-claim squared deviations tracked over T periods under a known deference path β_t , claim retirements observed, arrival cohorts observed with their masses. The natural estimators are

$$\hat{\lambda}_0 = \frac{\text{retirements}}{\text{claim-period exposure}}, \quad \hat{\eta} = -\text{slope of within-claim contraction ratios on } (1 - \beta_t),$$

$$(\hat{g}, \hat{I}_0) = \text{OLS slope and intercept of arrival mass on lagged stock},$$

the last because the content channel makes expected arrival mass affine in the lagged aggregate: $\mathbb{E}[\text{mass}_t \mid D_{t-1}] = I_0 + gD_{t-1}$.

Numerical Result 8.1 (Recovery and rates). The identification algebra above is exact – nonzero exposure recovers λ_0 from expected retirements, regressor variation recovers η and (g, I_0) from the

displayed regressions, and exact estimates reproduce the margin $v(\lambda_0 - g) - 2cI_0$ exactly; finite-sample accuracy is a battery measurement, reported here. In thirty seed-pinned battery panels ($T = 600$, three Gamma-cohort arrivals per period, sinusoidal β -path), mean errors are 2% for $\hat{\lambda}_0$, regression-exact for $\hat{\eta}$, 13% of the λ_0 scale for $\hat{g}$, and 3% for $\hat{I}_0$; the corresponding ninetieth percentiles are 4%, regression-exact, 26%, and 5%. In a separate rate experiment with twenty-four panels at each $T \in \{300, 600, 1200, 2400\}$, the fitted log-log RMSE slopes of the three noisy estimators ($\hat{\lambda}_0, \hat{g}, \hat{I}_0$) are $-0.61, -0.46, -0.41$; $\hat{\eta}$ is algebraically exact in that simulation, so its floating-point RMSE is not used as rate evidence. The plug-in margin $\hat{m} = v(\hat{\lambda}_0 - \hat{g}) - 2c\hat{I}_0$ predicted the true dynamical fate out of sample in 40/40 designed trials: estimated-certified loops were never captured from adversarial initial conditions, and estimated-capturable loops captured. These are descriptive frequencies under the stated simulation law and seeds, not distribution-free performance guarantees or field-population probabilities.

There is nevertheless an exact finite-sample sensitivity statement beneath these measurements. Write an observed retirement count as $\lambda_0 E + e_R$, two contraction moments as $1 - \eta r_j + e_j$, and two arrival-mass moments as $I_0 + gD_j + u_j$. For $E \neq 0$, $r_1 \neq r_2$, and $D_1 \neq D_2$,

$$\begin{aligned} \hat{\lambda}_0 - \lambda_0 &= e_R/E, & \hat{\eta} - \eta &= -\frac{e_2 - e_1}{r_2 - r_1}, \\ \hat{g} - g &= \frac{u_2 - u_1}{D_2 - D_1}, & \hat{I}_0 - I_0 &= u_1 - D_1 \frac{u_2 - u_1}{D_2 - D_1}. \end{aligned}$$

Thus the response and gain errors are bounded respectively by $(|e_1| + |e_2|)/|r_2 - r_1|$ and $(|u_1| + |u_2|)/|D_2 - D_1|$: insufficient excitation appears exactly as a small denominator, with no asymptotic shorthand. For fixed (v, c) , parameter errors $(\delta_\lambda, \delta_g, \delta_I)$ perturb the margin by exactly $v(\delta_\lambda - \delta_g) - 2c\delta_I$, whose magnitude is at most $|v|(|\delta_\lambda| + |\delta_g|) + 2|c||\delta_I|$. These identities and bounds are machine-checked in `PaperIII/FiniteSampleIdentification.lean`; they assume error bounds, not a sampling distribution, and therefore do not supply the empirical RMSEs.

For decisions, point estimates are not enough; the rule must be conservative against estimation noise on the dangerous side.

Proposition 8.2 (Margin variance and the safe-side implication). *Let the estimators be square-integrable, and let $\hat{\lambda}_0$ be independent of the joint OLS pair $(\hat{g}, \hat{I}_0)$ – the estimators use distinct data slices. Then the margin estimator $\hat{m} = v(\hat{\lambda}_0 - \hat{g}) - 2c\hat{I}_0$ satisfies, exactly,*

$$\text{Var}(\hat{m}) = v^2(\text{Var} \hat{\lambda}_0 + \text{Var} \hat{g}) + 4c^2 \text{Var} \hat{I}_0 + 4vc \text{Cov}(\hat{g}, \hat{I}_0).$$

Moreover the safe-side rule – certify iff $\hat{m} - 2\text{SE}(\hat{m}) > 0$ – is deterministically sound on its coverage event: whenever $|\hat{m} - m| \leq 2\text{SE}(\hat{m})$, certification implies $m > 0$.

Proof. Independence of $\hat{\lambda}_0$ from the joint pair makes $v\hat{\lambda}_0$ independent of $-v\hat{g} - 2c\hat{I}_0$ by measurable composition, so their variances add, while the covariance inside the pair remains; expanding $\text{Var}(v\hat{g} + 2c\hat{I}_0)$ produces the cross term. The implication is arithmetic: $m \geq \hat{m} - |\hat{m} - m| \geq \hat{m} - 2\text{SE}(\hat{m}) > 0$. $\square$

Five-parameter extension. When (v, c) are estimated rather than fixed, put $\vartheta := (v, c, \lambda_0, g, I_0)$ and let Σ be the joint covariance matrix of $\hat{\vartheta}$. The exact perturbation identity is

$$m(\vartheta + \Delta) - m(\vartheta) = \nabla m(\vartheta)^\top \Delta + \Delta_v(\Delta_{\lambda_0} - \Delta_g) - 2\Delta_c\Delta_{I_0},$$

with

$$\nabla m(\vartheta) = (\lambda_0 - g, -2I_0, v, -v, -2c).$$

Accordingly the full delta-method variance is

$$\text{Var}_\Delta(\hat{m}) = \nabla m(\vartheta)^\top \Sigma \nabla m(\vartheta). \quad (8.1)$$

The three-parameter identity in Proposition 8.2 is the exact fixed- (v, c) specialization under its stated slice-independence assumption. A sum of the five diagonal contributions is valid only under the additional condition that every off-diagonal covariance entering (8.1) vanishes; in particular, jointly estimated $(\hat{v}, \hat{c})$ and the OLS pair $(\hat{g}, \hat{I}_0)$ may not be treated as independent by default. The perturbation identity, positive-semidefinite covariance form, and diagonal specialization are machine-checked.

The safe-side implication is not tied to a normal approximation. For any registered radius R satisfying the coverage event $|\hat{m} - m| \leq R$, the decision $\hat{m} - R > 0$ implies $m > 0$ exactly. If $\hat{\vartheta} = \vartheta + \Delta$, the perturbation identity supplies the finite nonlinear radius

$$R_{\text{exact}}(\vartheta, \Delta) := |\nabla m(\vartheta)^\top \Delta| + |\Delta_v(\Delta_{\lambda_0} - \Delta_g) - 2\Delta_c \Delta_{I_0}|,$$

which always covers $|\hat{m} - m|$ by the triangle inequality. Thus a plug-in margin exceeding this linear-plus-remainder radius is a non-asymptotic certificate. The coverage-event implication, this exact radius, and the resulting certificate are machine-checked alongside the covariance algebra.

Coverage probability requires assumptions beyond the pointwise implication. A distribution-free finite-sample statement is available for a scalar margin estimator: if $\hat{m}$ is square-integrable and unbiased for m , with positive *true* variance σ^2 , then for every fixed $R > 0$,

$$\Pr\{|\hat{m} - m| > R\} \leq \frac{\sigma^2}{R^2}, \quad m \leq 0 \implies \Pr\{\hat{m} > R\} \leq \frac{\sigma^2}{R^2}.$$

Consequently $R = 2\sigma$ gives at least 75% coverage and at most 25% false certification without any distributional approximation. This deliberately weak Chebyshev calibration, including the false-certification transfer, is machine-checked in `PaperIII/CoverageProbability.lean`. It uses the true standard deviation and unbiasedness; it does not justify replacing σ by a random estimated standard error.

For a finite registered family of primitive tests, joint coverage is a separate obligation. If test j has failure event F_j and a proved marginal bound $\Pr(F_j) \leq \delta_j$, then $\Pr(\bigcup_j F_j) \leq \sum_j \delta_j$; any false composite verdict that is possible only when some primitive event fails inherits the same sum. This Bonferroni transfer is exact and requires no independence, and is also machine-checked in `PaperIII/CoverageProbability.lean`. Thus a protocol must allocate a familywise error budget; repeated use of an unadjusted “two-SE” radius does not by itself calibrate the composite verdict.

In operation the variances are estimated rather than known: $\widehat{\text{Var}}\hat{\lambda}_0 = \hat{\lambda}_0(1 - \hat{\lambda}_0)/\text{exposure}$ under a Bernoulli retirement model at fixed exposure, and the OLS covariance estimator for the pair. Under the additional asymptotic assumptions of a limiting standard normal law for $(\hat{m} - m)/\widehat{\text{SE}}(\hat{m})$ and consistent standard-error estimation, the rule’s one-sided false-certification probability tends to $\Phi(-2) \approx 2.3\%$ at the boundary $m = 0$ and is no larger for fixed $m < 0$. The variance identities and coverage-event implications above are exact; this estimated-SE calibration is asymptotic.

Numerical Result 8.3. In forty seed-pinned battery draws the safe-side rule produced zero false certifications across all capturable loops (0/18) while retaining power 20/22 on comfortably certified loops. The event $|\hat{m} - m| \leq 2\widehat{\text{SE}}(\hat{m})$ occurred in 37/40 draws, an empirical coverage frequency of

92.5% with a 95% Wilson interval [80.1%, 97.4%]. This interval estimates coverage only under the battery’s simulation law; it is neither the finite-sample Chebyshev theorem (which uses true variance) nor proof of the asymptotic normal calibration.

The pipeline is therefore closed: observable panels $\rightarrow$ loop-parameter estimates $\rightarrow$ margin with standard errors $\rightarrow$ a certification decision whose error structure is one-sided in the safe direction.

9 Regulatory translation

The deliverable is a menu of inequalities on loop properties, each priced in margin units $m = v(\lambda_0 - g) - 2cI$, none of which mentions the knob:

Gain caps. Require $g \leq \bar{g} < \lambda_0$ (per-community $\sum_j g_{kj} < \lambda_{0k}$ in networks, or the sharp spectral form $\rho(\bar{S} + G) < 1$). The certificate curve of Figure 2 is the price schedule: the usefulness premium needed for guaranteed safety is $2cI/(\lambda_0 - \bar{g})$, diverging as the cap approaches the grounding rate. Content-ranking and recommendation choices that determine how strongly standing disagreement spawns fresh contested claims are the physical lever.

Grounding floors. Require $\lambda_0 \geq \underline{\lambda}$: institutional resolution capacity – fact-checking with reach, adjudication, authoritative settlement – raises the divergence wall and cheapens every other instrument simultaneously, since both zone boundaries scale as $(\lambda_0 - g)^{-1}$.

Audit-sensor mandates. Require an unweighted-error sensor of weight ε chosen from a latency target via Remark 6.5. By Theorem 6.1 this is the menu’s *only* instrument that operates after the certificate fails: the satisfaction channel at capture carries zero information about the diverging coordinate, so within the model no analysis of satisfaction telemetry, however sophisticated, can substitute.

Cadence caps. Bound the preference-update rate relative to belief mixing: by the separatrix limit (Proposition 7.6), the protected shock size is $\kappa^*(\beta_0)/\alpha$ to leading order, so halving cadence asymptotically doubles the transient excess stock the loop can absorb without capture. Pricing the cap at fixed operating scale needs an explicit separatrix-rate envelope – open in Section 11; the envelope-to-bounds transfer is machine-checked in `PaperIII/CadenceRate.lean`.

A regulator certifies the loop the product runs inside; the product’s behavior – the knob – is never the regulated quantity. By Theorem 4.3 it is also robust to the product as the knob cannot be, the certificate binding every sign-consistent responsive rule at once.

This menu is deliberately unlike the instruments now in use, and the difference is the object rather than the rigor. Capability evaluation and deployment gating test a system before release (Shevlane et al., 2023); sociotechnical and holistic evaluation widen the unit of assessment to deployed impact (Weidinger et al., 2023; Solaiman et al., 2025); algorithmic auditing examines a deployed system from outside (Metaxa et al., 2021), or through an internal accountability process (Raji et al., 2020). Each takes the system’s conduct as the object, and each therefore has to be repeated whenever the conduct changes. The inequalities above take the loop as the object and are for that reason indifferent to the conduct. The audit-sensor mandate is where the two traditions meet: it is an instrumentation requirement of exactly the kind auditing already asks for, except that its necessity here is a theorem – Theorem 6.1 – rather than a matter of good practice, and its sufficient magnitude is set by a latency target rather than negotiated. The measurement side of that meeting – estimands, experimental arms, and decision thresholds for locating a deployed system on the pressure–drift–capture ladder, its certified core built from the sub-saddle trap at its explicit constants – is pre-registered in the companion audit protocol (Hofheinz, 2026a).

10 Verification

Every formal claim above is paired with an independent numerical witness in a seed-pinned battery of fifteen suites (`derisk_loops.py`, seed 23), written before the corresponding prose. The suites verify, in order: the closed stationary forms of all four conservation cells against long iteration and the conservation loci ($\zeta = 0$; $g = \lambda_0$) including the interior-minimum location u_c (N0a–c); the certificate’s $G_g < 0$ conclusion and the calibration of four adaptation rules – gradient, sign, momentum, multiplicative – from three adversarial initial conditions on sixty certified draws, the local stability of the captured corner below the capture line on sixty draws, and the intermediate zone’s dichotomy including machine-precision convergence to interior partial-capture equilibria (N1a–c); both gain ceilings with horizons scaled to the spectral gap (N2); the observability ranks $1/2/2$, the kernel membership of the unstable mode, and the latency law with fitted slope 41 against predicted 42 (N3); the monotone growth of the separatrix with falling α (N4); panel recovery of $(\lambda_0, \eta, g, I_0)$ and the 40/40 out-of-sample fate predictions (N5); the jump-map limit κ^* , the integral approximation K , the six-point product table with Richardson extrapolation within 0.9%, and the near-saddle continuum agreement within 1% (N6); the exact nonlinear indistinguishability – identical satisfaction outputs to the last bit over four hundred steps – and the sensor-gap growth rate within 0.4% of $2\ln(1 + \zeta)$ (N7); the $T^{-1/2}$ estimator rates and the safe-side rule’s 0/18 false certifications with 20/22 power (N8); the network bound, the multi-knob certificate, the stability ceiling on both sides, and the exact scalar recovery (N9); and the burn-decomposition identity to machine precision, the two-sided K -bounds with strictly decreasing jumps, the near-saddle ratio, the trichotomy of Proposition 7.4(ii) – strict increase below the clipping plateau, $\beta_\infty \equiv 1$ on it, and the plateau strictly above κ^* at $\beta_0 = 0, 0.4, 0.9$ (occupying 35.1%, 60.7%, 95.9% of the admissible interval) – characteristic capture, and the upper half-limit at small α (N10). The battery has the standing to falsify, not merely to confirm: each suite asserts a claim, so a failing suite is a refutation rather than a defect report. The interior attractors of Theorem 4.5(B) and the separatrix scale of Remark 7.2 are constrained by the suites independently of their derivations. The battery is the method, not an appendix to it.

Every theorem, lemma, proposition, and corollary of this paper is additionally machine-checked in Lean 4 against Mathlib, with no unproved steps and no axioms beyond Lean’s standard three; the numerical results are battery measurements and lie outside the formalization by type. The development resides at `lean/` in the replication repository, and `#print axioms` reports the exact dependency set of any declaration. Sharpness is machine-checked in the same development: the clipped-plateau witnesses printed in Proposition 7.4(ii), the projection non-removability of Theorem 7.1, and the companion’s period-two instance behind the ascending-branch proviso of Theorem 2.2 witness that these hypotheses are necessary rather than conventional.

11 Open problems

Endogenous gain. Here g is a fixed property of the content channel. In a content market, g itself responds to engagement – polarizing claims are supplied because they perform – creating a loop around the loop whose fixed points need not respect any static cap. The natural conjecture is that the certificate survives with g replaced by its market equilibrium value, but the equilibrium may sit on the wrong side of the wall.

Welfare foundations for v . The certificate prices safety in units of the usefulness premium; deriving v from primitives – what users would pay for grounded answers – would convert the certificate from a consistency condition into a welfare statement.

Partial-capture economics. Theorem 3.1 gives operators with moderate supercritical gain a stationary *incentive* to stop at β_c , and Theorem 4.5 gives the matching attractor. Whether competition among operators selects for or against the partial-capture regime is open, and is the regime most likely to be observed in the wild precisely because no alarm fires there.

Network identification. Section 8 identifies a scalar gain; identifying the coupling matrix G – and hence the Perron margin $1 - \rho(\bar{S} + G)$ – from cross-community panels is the spectral analogue, presumably via the lag structure of cross-community arrival masses.

Sharper pricing of κ^ .* Two-sided refinements of Proposition 7.4(iii) at fixed β_0 , rather than asymptotically, and explicit rates in α for the separatrix limit of Proposition 7.6, would price the cadence cap exactly at operating scale.

Stochastic transit. The companion’s Arrhenius protection concerns noise-driven first saddle-policy crossing, the proved precursor of basin exit; the jump-map limit concerns deterministic transit across a basin. Their composition – noisy adaptation at small α crossing the jump-map separatrix – is the realistic regime and should interpolate between the two scalings.

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